Vipul Sapra

Charlotte, North Carolina | 513-693-6897 | vpsapra5@gmail.com | linkedin.com/in/vipul-sapra-ab30a09a

EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Generative AI, Agentic Workflows, LLM Orchestration, GitHub Copilot Integration, Autonomous AI Agents, RAG & Vector Embeddings, Predictive Machine Learning, MLOps, Time-Series Forecasting (SARIMA), Data Governance & Lineage, Dimensional Modeling (Star/Snowflake, SCD Type 1/2), Automated Testing & Evaluation Frameworks, Executive Decision Intelligence

Programming Languages & Tools: GitHub Copilot, KIRO IDE, Azure (ADF, Databricks, Synapse, Azure SQL), AWS (S3, Redshift), Snowflake, Apache Spark, PySpark, Power BI (DAX), Tableau, Python, Scikit-Learn, XGBoost, Collibra

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data & AI Systems Engineer - People Analytics & Intelligent Platforms — **May 2022 – Present**

- **Engineered custom autonomous agentic workflows and prompt orchestration** utilizing KIRO IDE and GitHub Copilot to automate schema discovery, regex classification, and code migration, reducing manual delivery turnaround by 40-50% while empowering teams with self-serve analytical capabilities.
- **Built automated testing and evaluation frameworks for conversational AI** platforms, cutting manual QA turnaround by 50% and accelerating production release cycles for enterprise virtual assistants and self-serve tooling.
- **Developed automated data quality firewalls and source-to-target reconciliation** pipelines integrated with Collibra data lineage, ensuring strict data governance, privacy compliance, and trustworthy data handling across high-throughput cloud environments.
- **Architected distributed lakehouse pipelines and dimensional models** using Azure (ADF, Databricks, Synapse, SQL) and Apache Spark / PySpark to crunch hundreds of terabytes with 99.9% availability SLAs, feeding reliable downstream analytical engines.
- **Translated complex telemetry into executive decision intelligence** by constructing Key Operating Metric (KOM) semantic models and dashboards in Power BI (DAX) and Tableau that guide enterprise operational strategy.

Data Scientist & Machine Learning Specialist — **May 2022 – Present**

- **Engineered autonomous machine learning anomaly detection frameworks** identifying schema variations, distribution drift, and ETL discrepancies with 95% precision to safeguard downstream BI integrity.
- **Built and deployed scalable predictive classification models** using XGBoost, Random Forests, and Scikit-Learn to analyze customer behavior and deliver actionable behavioral signals improving service reliability by 20%.
- **Architected advanced time-series forecasting models (SARIMA)** to predict field support demand and truck rolls, optimizing workforce capacity allocation and reducing response times by 20%.

Business Intelligence & Decision Analytics Specialist — **May 2022 – Present**

- **Designed enterprise-grade dimensional models and data warehouses** using Star/Snowflake schemas and SCD Type 1/2 methodologies on AWS (S3, Redshift) and Snowflake platforms.
- **Developed semantic layers and executive reporting pipelines** translating transactional telemetry into high-impact operational metrics for corporate stakeholders.
- **Automated data reconciliation processes** to eliminate manual reporting overhead and reduce operational turnaround times by over 80%.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.